**Kafrelsheikh University - Faculty of Medicine**
**Physiology (practical) for first level students (5+2) Program**
**Date:** 31/12/2024
**Time allowed:** 30 minutes **Student name/** .....
**Total marks:** 20 marks **Student number/** .....

Answer the following questions (All questions to be answered):

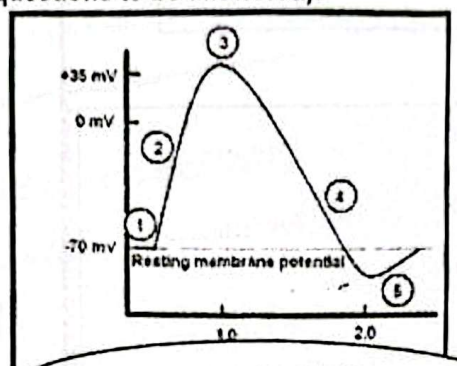


Figure 1 action potential

**In figure 1, what process is responsible for the change in membrane potential that occurs between point 1 and point 3?**

A. Movement of Na+ into the cell
B. Movement of Na+ out of the cell
C. Movement of K+ into the cell
D. Movement of K+ out the cell

**In figure 1, what process is responsible for the change in membrane potential that occurs between point 3 and point 4?**

A. Movement of Na+ into the cell
B. Movement of Na+ out the cell
C. Movement of K+ into the cell
D. Movement of K+ out of the cell

**In figure 1, what process is responsible for the change in membrane potential that occurs at area 5?**

A. Movement of Na+ into the cell
B. Movement of Na+ out the cell
C. Activation of the Na+ - K+ pump
D. Inhibition of the Na+ - K+ pump

**In figure 1, at which labeled point on the action potential the absolute refractory period is located?**

A. 5
B. 2 & 3.
C. 4.
D. 4 & 5



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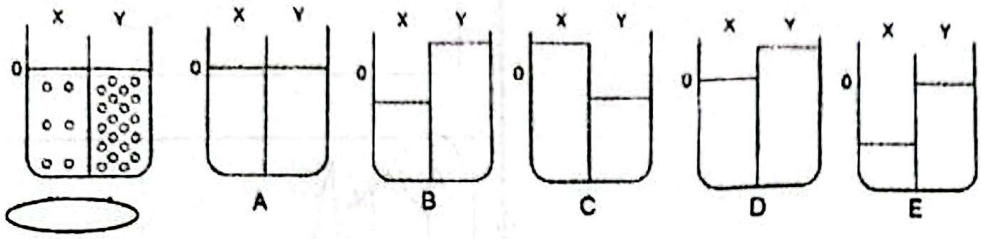
**In the figure 2, two compartments (X and Y) are separated by a typical biological membrane (lipid bilayer). The concentrations of glucose in compartments X and Y at time zero are shown. There are no transporters for glucose in the membrane, and the membrane is impermeable to glucose. Which of the figures best represent the volumes of compartments X and Y when the system reaches equilibrium?**




**Which of the figures best represent the volumes of compartments X and Y when the system reaches equilibrium?**

A. A B. B C. C D. D

**Facilitated diffusion differs from simple diffusion because it:**

A. is according to the concentration gradient B. Is a passive transport

C. Involves a carrier protein D. Involves expenditure of energy

**In primary active transport energy is derived from:**

A. ATP breakdown B. Ionic concentration differences across two sides of cell membrane

C. Golgi apparatus D. Counter transport of calcium and hydrogen ions

**Which fluid compartment contains about Two third (by volume) of all body water?**

A. intracellular fluid B. plasma

C. lymph D. extracellular fluid

**About what percent of a young adult's body weight is due to water?**

A. 4 to 5% B. 10 to 12% C. 60 to 65% D. 98 to 99%

**The major cation in the extracellular fluid is**

A. Na+ B. K+ C. Cl - . D. Ca2+

**The major cation in the intracellular fluid is**

A. Na+ B. K+ C. Cl- D. Ca2+

**Osmosis is the diffusion of _____ across a selectively permeable membrane.**

A. Urea B. Oxygen

C. Water D. Sodium

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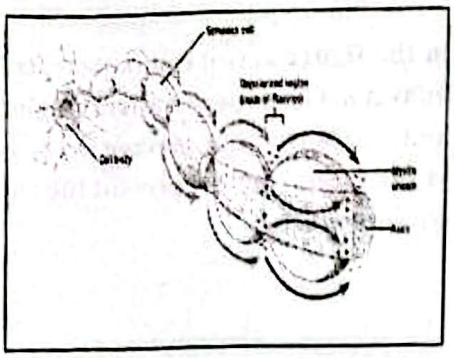


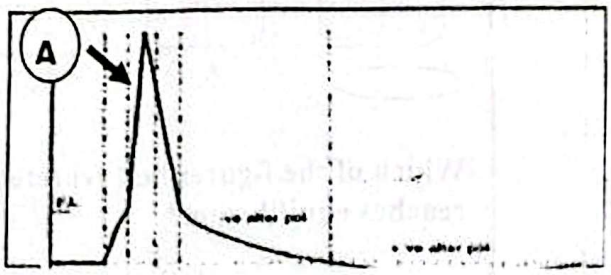
**This type of conduction of action potential:**

a) Occurs in unmyelinated nerve fibres.

b) Is slower than the point to point conduction.

c) Consumes less energy than point to point conduction.

d) None of the above.



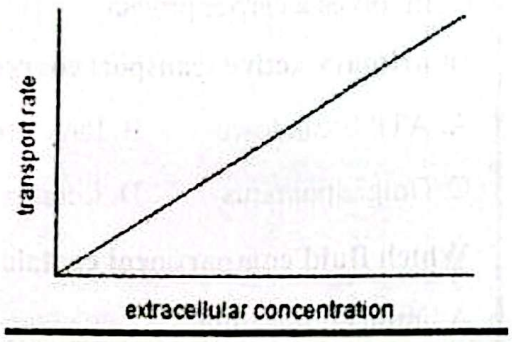

**What is the period A?**

A) Relative refractory period

B) Absolute refractory period

C) Super normal period

D) Sub normal period



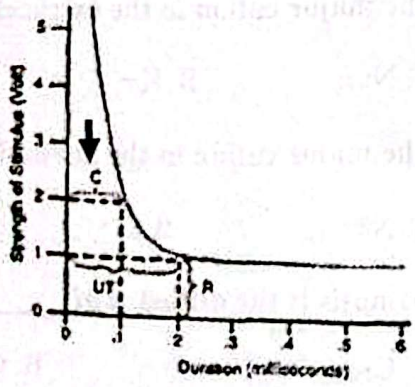

**While studying the effect of varying the extracellular concentration on the transport rate of a substance, the graph was obtained. This finding would suggest that this substance is transported by:**

A. Simple diffusion

B. Ion-selective channels

C. Facilitated diffusion

D. Active transport

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. a) Is the threshold stimulus

. b) Is twice the rheobase

.c) Can be used as a measure of excitability

d) Is the time needed to excite a nerve by a current strength equal to the rheobase.



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